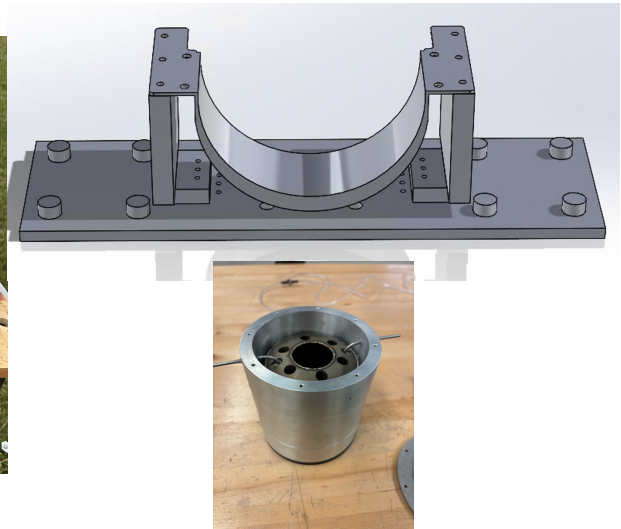
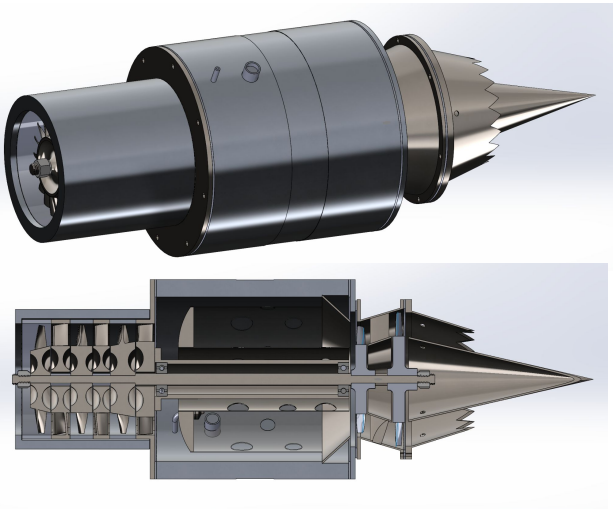


JASON SIMON

MECHANICAL ENGINEERING AT THE UNIVERSITY OF PENNSYLVANIA

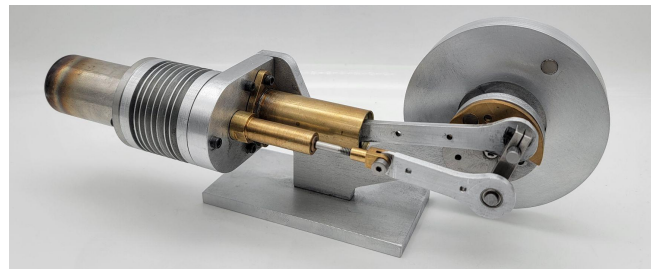
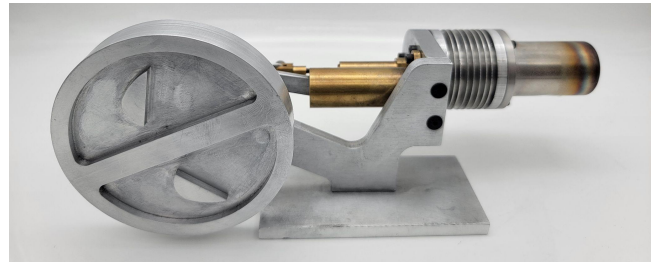
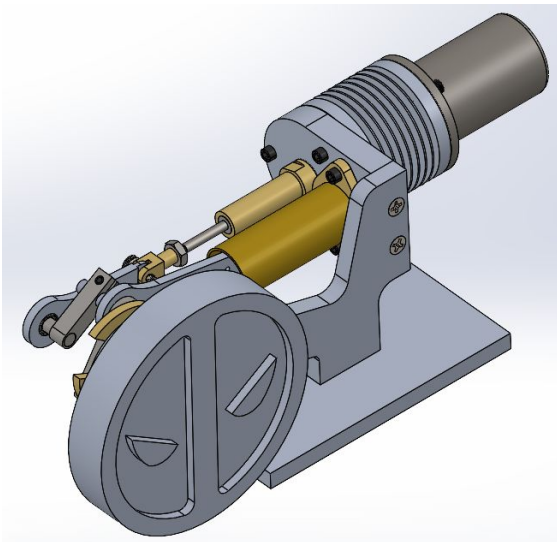
 jtsimon@seas.upenn.edu
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PENN JET PROPULSION



- Part of a team creating the first functioning micro turbojet engine fully developed from scratch by undergraduate students
- Engine achieved sustained combustion during testing and has a theoretical thrust of 140N and thrust to weight ratio of 3.5
- Designed in SolidWorks and manufactured using a ProtoTrak mill and lathe a new mount to hold the engine securely during future testing
- Worked with 2 other students to design the combustion chamber and plan out the overarching integration and interfacing of all the subsystems

STIRLING HEAT ENGINE



- Designed and fabricated a stirling heat engine with a Deadpool inspired design from scratch using Prototrak mill, lathe, bandsaw, and CNC operations for metal machining
- Modeled and assembled all parts in SolidWorks with mates to replicate final engine movements
- Created ASME standard engineering drawings to obtain machining approval for all custom parts
- Flywheel reached 1350+ RPM during testing

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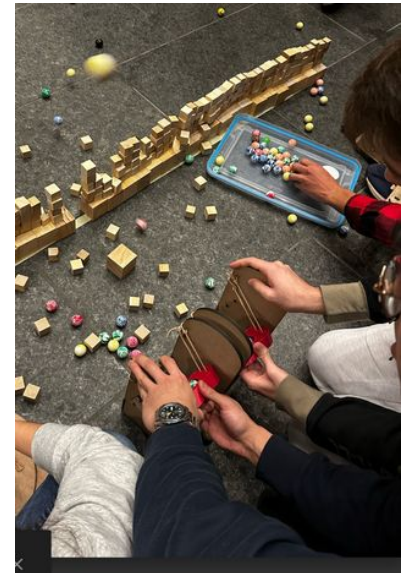
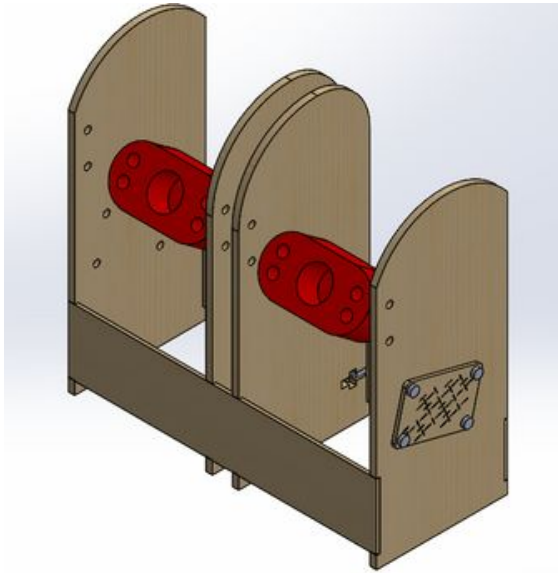


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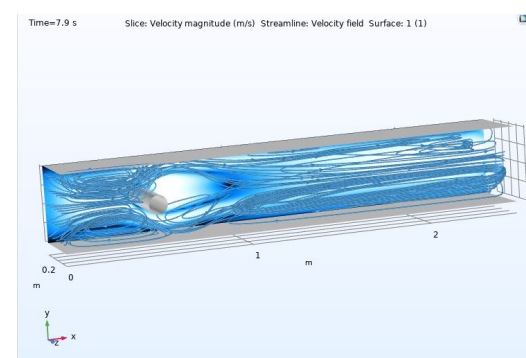
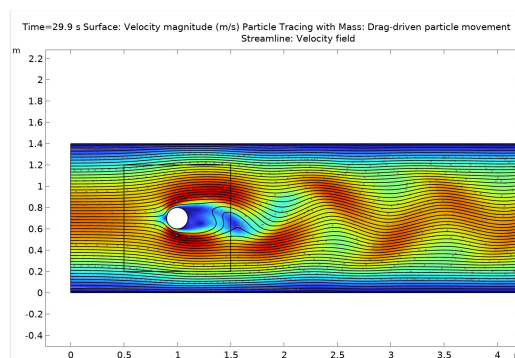
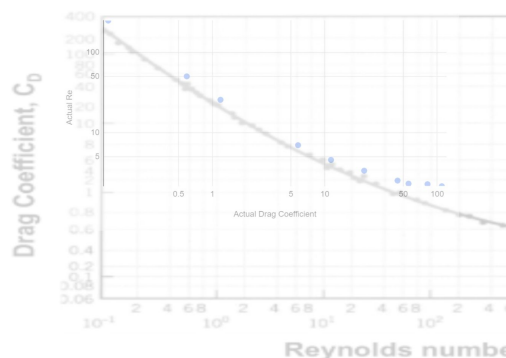
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SIEGE MACHINE



- Worked in a team of 3 to design (using SolidWorks for CAD modeling) and manufacture a siege machine to launch bouncy balls using only laser cutting, 3D printing, rubber bands, and fasteners
- Laser cut and assembled as many parts as possible with press fits for easy disassembly to facilitate rapid iteration as we tested it
- Produced a dual shooter slingshot mechanism after testing various other prototypes including a pinball machine mechanism and a catapult to create the most effective siege machine possible with the given constraints

COMSOL FLUID DYNAMICS MODELING



- Modeled fluid flow over different shapes using COMSOL in both 2D and 3D in order to calculate the drag coefficient of different shapes
- Simulation is planned to be used to optimize cars, planes, and more projects for different clubs at Penn in the future